Fabric Defect Classification

End-to-end comparison of Roboflow DINOv3 and locally trained ResNet18

1. Objective

The goal was to classify fabric images into two categories: **defect_free** and **minor_defect**. Three models were evaluated: an original Roboflow-hosted DINOv3 model, an augmented DINOv3 model, and a locally trained ResNet18.

2. Dataset and preparation

The Roboflow dataset contained 5,000 images and used a 70/20/10 train/validation/test split. Images were auto-oriented and resized to 224 × 224 pixels. The augmented Roboflow version applied horizontal flipping and random rotation from -10° to +10°, with a 3× augmentation multiplier.

The local ResNet18 folders reported 3,553 training images, 1,015 validation images, and 509 test images (5,077 total). The full evaluation compared predictions for the same 509 test filenames and actual labels across all three model outputs.

Dataset consistency limitation

Roboflow's dataset version contains 5,000 images, while the local ResNet18 folders total 5,077. Although both evaluations used 509 test images, the training and validation counts differ. It has not been confirmed that the local ResNet18 training/validation images exactly match the Roboflow split. This should be disclosed when presenting the comparison.

3. Model results

Model	Accuracy	Macro F1	Correct / 509
Original DINOv3 (v1)	41.65%	40.71%	212
Augmented DINOv3 (v2)	60.51%	60.24%	308
ResNet18 (CPU)	82.32%	79.49%	419

The augmented DINOv3 model improved accuracy by 18.86 percentage points over the original DINOv3 model. ResNet18 achieved the highest accuracy (82.32%) and macro F1 (79.49%) on the 509-image evaluation set.

ResNet18 class-level results

Class	Precision	Recall	F1
defect_free	63.19%	83.33%	71.88%
minor_defect	92.97%	81.94%	87.11%

4. ResNet18 confusion matrix

Actual / predicted	defect_free	minor_defect
defect_free	115	23
minor_defect	67	304

Rows are actual classes and columns are predicted classes. The class order is defect_free followed by minor_defect.

5. Failure analysis

ResNet18 missed 67 defective images by predicting them as defect-free (false negatives). It also incorrectly flagged 23 defect-free images as defective (false positives). Missed defects matter in a quality-inspection setting because a faulty fabric may pass inspection.

Possible contributors include small or subtle defects, low contrast, lighting variation, and complex fabric patterns. These are hypotheses, not confirmed causes; a visual review of the misclassified images would be needed to establish the reasons.

6. Inference performance

The local ResNet18 benchmark processed 509 test images in 56.581 seconds, averaging 111.16 milliseconds per image on CPU. Original DINOv3 took an average of 1,343.43 ms per image and augmented DINOv3 took 1,235.48 ms per image. The DINOv3 values are end-to-end API request times that include network communication, so they are not directly equivalent to local CPU inference.

7. Training time and limitations

The exact ResNet18 training duration was not recorded. It is reported as **not recorded**, rather than estimated. The training/validation split-count difference between Roboflow and the local folders is another limitation.

8. Conclusion

ResNet18 produced the strongest measured results on the 509-image test set, with 82.32% accuracy and 79.49% macro F1. Roboflow augmentation improved DINOv3 accuracy from 41.65% to 60.51%, but the augmented model remained below ResNet18 in this evaluation. Future work should confirm identical data splits, record training duration automatically, and inspect false-negative images to reduce missed defects.

Environment note: Python 3.11.9 and PyTorch 2.15.0+cpu were recorded. Add permitted scripts and model weights to the repository separately. Never upload API keys or credentials.